

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

March-April 2018

CL 315 Basic Transportation Systems

Date: 27.03.2018, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 State whether the following statements are True or False: [07]

- 1) Airways are the fastest mode of transportation.
- 2) The gauge of the railway track is 0.75 times the diameter of wheel of train.
- 3) Presently, Indian railway is divided into 20 administrative zones.
- 4) Group C Lines of Indian Railway system is classified as the routes of sub-urban areas.
- 5) Meter Gauge has a spacing of 1.1 m between the two rails.
- 6) The function of sub-ballast layer is to provide effective drainage to the track.
- 7) Tongue rails are the weakest link in the permanent way.

Q - 2 (a) Explain superelevation with neat sketch. [05]

OR

- (a) Define Gradient. How it is measured? Explain the types of gradients. [05]
- (b) Compare Railway and Roadway. [05]
- (c) Why transition curves are provided on railway tracks? [04]

Q - 3 (a) Draw a neat sketch of cross-section of a B.G. Track for a double line on straight railway track. [05]

(b) Define the following terms of Railway Engineering: [05]

- (1) Crossings (2) Turnouts (3) Point and Switches (4) Platform (5) Yard

OR

- (b) Draw neat sketch of following track junctions given below: [05]
- 1) Scissor Cross-over
 - 2) Gathering Lines
 - 3) Temporary Diversion
 - 4) Triangle
 - 5) Double Junctions
- (c) Why signals are necessary to be provided in railways? [04]

SECTION – II

- Q - 4 Define the following terms used in Docks and Harbour: [07]
- (1) Breakwater (2) Jetties (3) Pier (4) Transit Shed
(5) Light house (6) Fenders (7) Dolphin
- Q - 5 (a) Write a critical note on Water Transportation. [05]

OR

- Q – 5 (a) List out the factors consider in designing of harbour. Also enlist ship characteristics. [05]
- (b) Explain different aircraft wheel configurations with neat sketch. [05]
- (c) Enlist the key functions of airport authority of India. [04]
- Q-6 (a) How site selection of airport is carried out? [05]
- (b) Explain Wind Rose Diagram with neat sketch. [05]

OR

- (b) Draw different types of runway patterns. What basic corrections are required to be carried out to the length of runway during the designing? [05]
- (c) What are the needs of air control aids? [04]
